

Case Study VI: OECD

By

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Executive Summary

This project analyzed migration skill flows, labor-market integration, and innovation patterns in Australia, Greece, and Norway from 2000–2020. The goal was to understand how skilled migration contributes to economic inclusion and national innovation outcomes within OECD countries. Using OECD datasets, including the DIOC, labor-market indicators, and R&D expenditure data, the analysis standardized education levels, compared immigrant and native employment trends, and assessed the relationship between high-skill migration and R&D investment. The findings show that high-skill migration increased across all three countries, with Australia experiencing the largest growth, Greece continuing to receive mainly low and medium-skill migrants, and Norway showing steady increases in high-skill inflows. Immigrants in Australia and Norway became more educated than natives, while Greece displayed more mixed education patterns. Labor-market outcomes improved broadly, with higher immigrant employment rates and narrowing unemployment gaps over time. Innovation analysis revealed a strong correlation between high-skill migration and R&D investment in Greece ($r = 0.951$) and Norway ($r = 0.917$), and a positive but weaker relationship in Australia ($r = 0.757$). These results are shaped by data inconsistencies, the limited country scope, and the broad nature of the indicators used. Recommendations include simplifying high-skill migration pathways, strengthening integration supports, aligning migration and innovation policies, and improving OECD data quality to enhance future analysis.

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1. Introduction

1.1 Background

The Organization for Economic Co-Operation and Development (OECD) is a global forum for policy analysis and collaboration. Its member countries face demographic challenges such as aging populations and shrinking labor forces. Skilled migration is often viewed as a solution, but outcomes depend on how effectively migrants are integrated into labor markets and innovation systems.

1.2 Objectives

This study examines migration skill flows and economic inclusion in the OECD countries Australia, Norway, and Greece between 2000 and 2024. Analysis in this project is based on OECD datasets such as the Database on Immigrants in OECD and non-OECD Countries (DIOC), R&D statistics, and labor market indicators.

2. Methods

2.1 Country Selection

Australia, Greece, and Norway were selected for the project because they represent distinct migration contexts within the OECD. Australia is a high-immigration country with long standing skill-based entry systems, Greece has experienced substantial migration pressures shaped by economic volatility, and Norway reflects a high-income welfare state with structured labor-market institutions. Examining these three countries together captures different policy environments, economic conditions, and challenges relevant to migrant labor-market outcomes.

2.2 Employment and Education

Labor-force data from multiple DIOC years were cleaned and standardized to align variable definitions, then combined into a panel for Australia, Greece, and Norway. Aggregated population counts were used to calculate employment and unemployment rates for immigrants and natives following OECD and ILO standards. Time-series plots were generated to compare labor-market indicators across countries and over years.

Employment and unemployment rates were analyzed for immigrant and native populations because they provide direct measures of labor-market integration. Differences in these rates reflect how immigrants participate in the labor force, access employment opportunities, and face barriers relative to native populations. These metrics are widely used in OECD labor-market evaluation, are comparable across different countries and time, and offer a way for assessing economic inclusion and potential mismatches in the utilization of migrant human capital.

Examining education levels is important because the study focuses on how the skill profile of migration relates to labor-market inclusion and broader economic performance. Education is one of the clearest indicators of workforce capabilities, shaping individuals' employment prospects and the types of jobs they are likely to access. Comparing the educational distributions of immigrants and natives helps show whether migrants enter with similar, higher, or lower levels of training, and whether host countries are making effective use of the skills they bring. These patterns also help contextualize differences in employment and unemployment rates and provide insight into how well each country is integrating and utilizing its migrant workforce.

2.3 Skills Migration

To evaluate how the skill composition of migrant inflows evolved over time, the DIOC datasets were processed, and all education variables were standardized into harmonized low, medium, and high-skill categories. This standardization enabled the calculation of skill shares for each country and year, forming a consistent basis for comparison. Stacked bar charts and line graphs were employed because they provide a clear representation of structural changes in migrant skill profiles, directly supporting the project's objective of identifying long-term shifts in human-capital inflows and assessing their relevance for labor-market inclusion outcomes. These methods facilitate precise interpretation of whether destination countries have strengthened or diminished their ability to attract highly educated migrants, which is essential for addressing the study's first guiding question on the evolution of skill-specific migration flows.

2.4 R&D and Innovation

To examine the relationship between skilled migration and national innovation capacity, tertiary-educated migrant shares were merged with OECD data on gross domestic expenditure on research and development. Visualizations were used to present these indicators, and this approach addresses the study's objective of determining whether countries with stronger innovation ecosystems exhibit migration patterns that promote enhanced economic performance. By aligning high-skill inflow trends with R&D investment intensity, this method provides a good way to evaluate the compatibility between human-capital attraction and national innovation strategies, addressing the guiding question regarding the connection between skilled migration and innovation outcomes.

3. Results

3.1 OECD Skill Migration Trends

Figure 1 shows the migration inflows by skill level across all OECD countries between 2000 and 2020. High-skill migration increases slightly, especially in 2020.

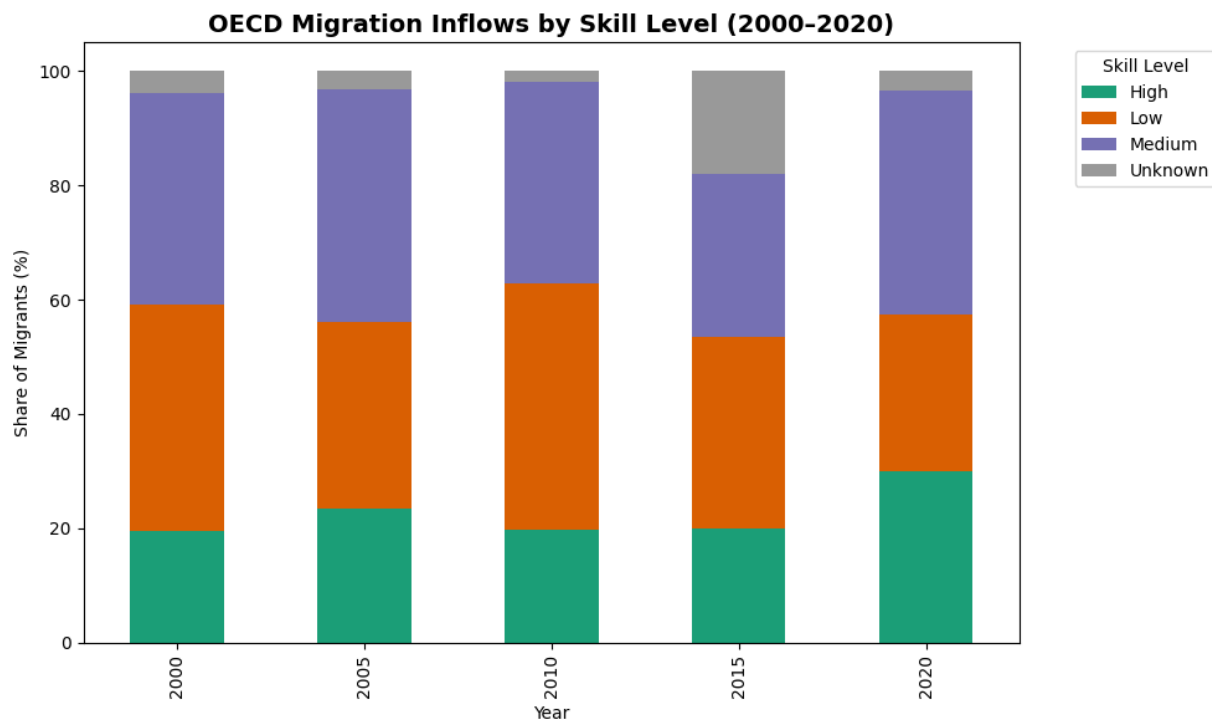


Figure 1 Migration inflows by skill level in all OECD countries from 2000-2020.

Figure 2 and Figure 3 present two versions of the migration inflows by skill level in Australia, Greece, and Norway. Australia experienced a consistent growth in high-skill and medium-skill inflows, and a decrease in low-skill especially by 2020. Greece experienced a slight increase in high-skill inflows, but low-skill and medium-skill inflows still outnumber high-skill. Norway's low-skill migration was highest around 2010 and 2015, but high-skill inflow began to increase around 2015.

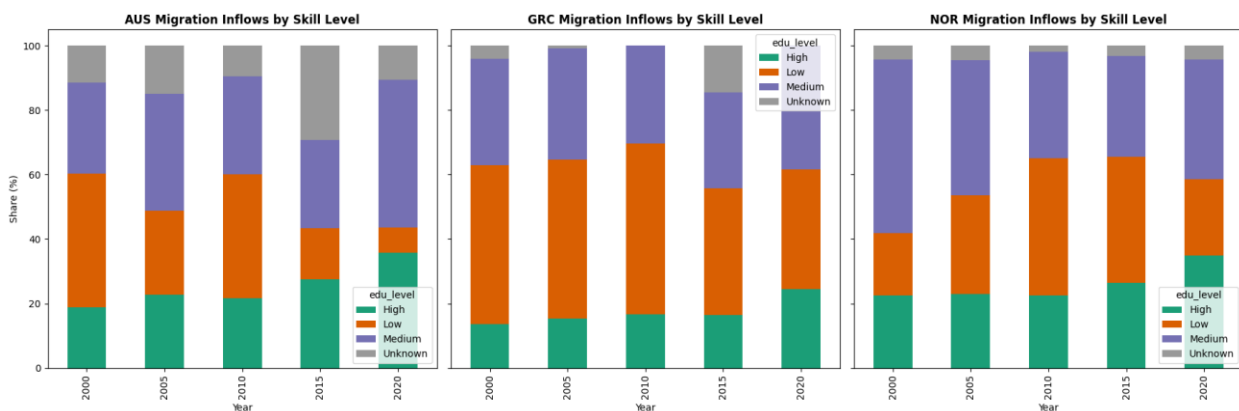


Figure 2 Migration inflows by skill level from 2000-2020 in Australia, Greece, and Norway.

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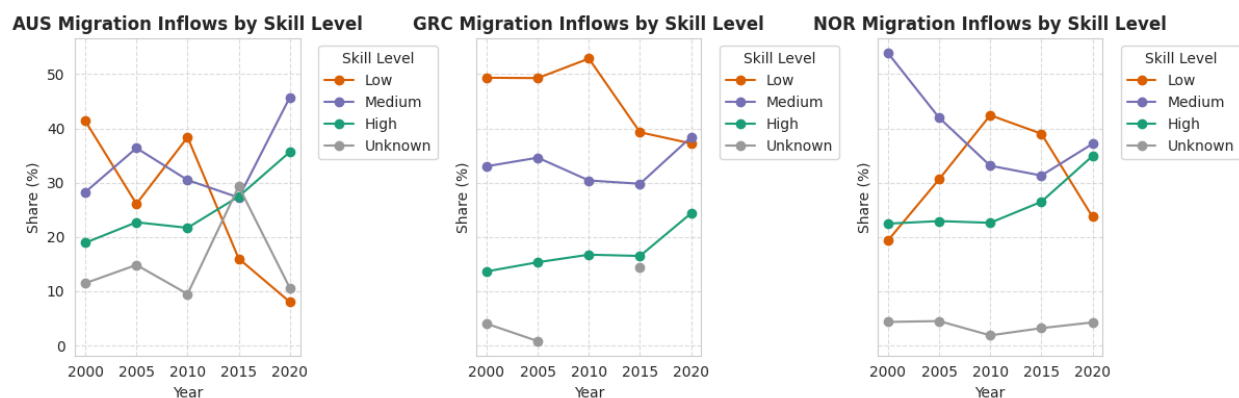


Figure 3 Line plots of migration inflows by skill level from 2000-2020 in Australia, Greece, and Norway.

Figure 4 and Figure 5 show the total migration into Australia, Greece, and Norway and the share of high-skill migrants in each of the countries.

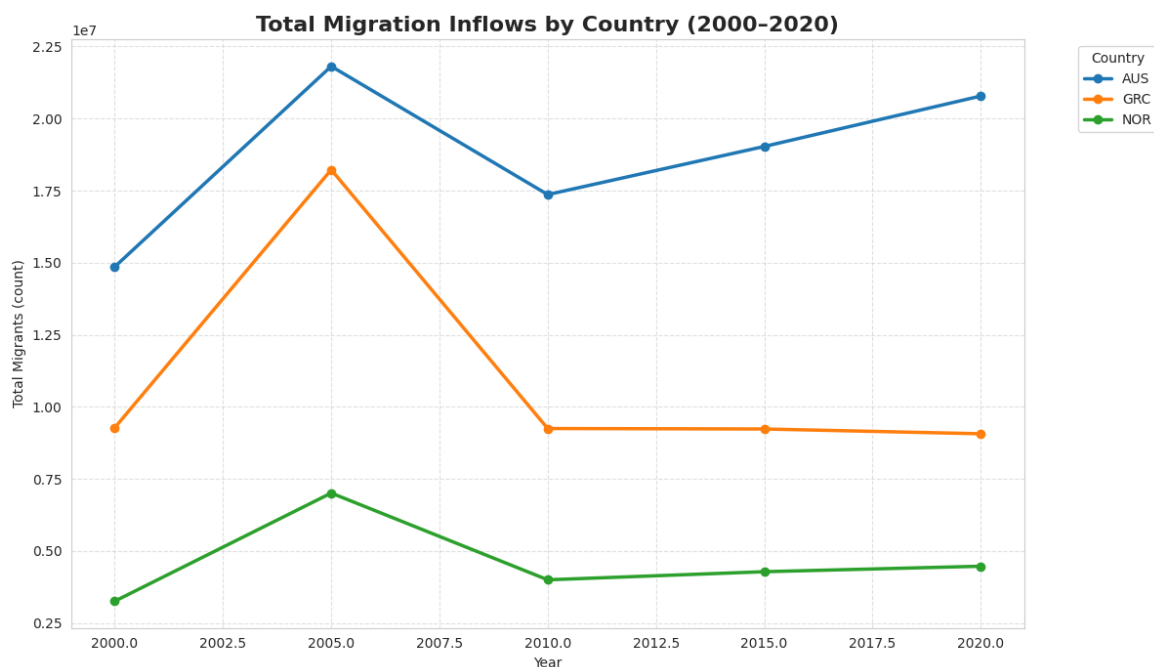


Figure 4 Total migration inflows by country in Australia, Greece, and Norway.

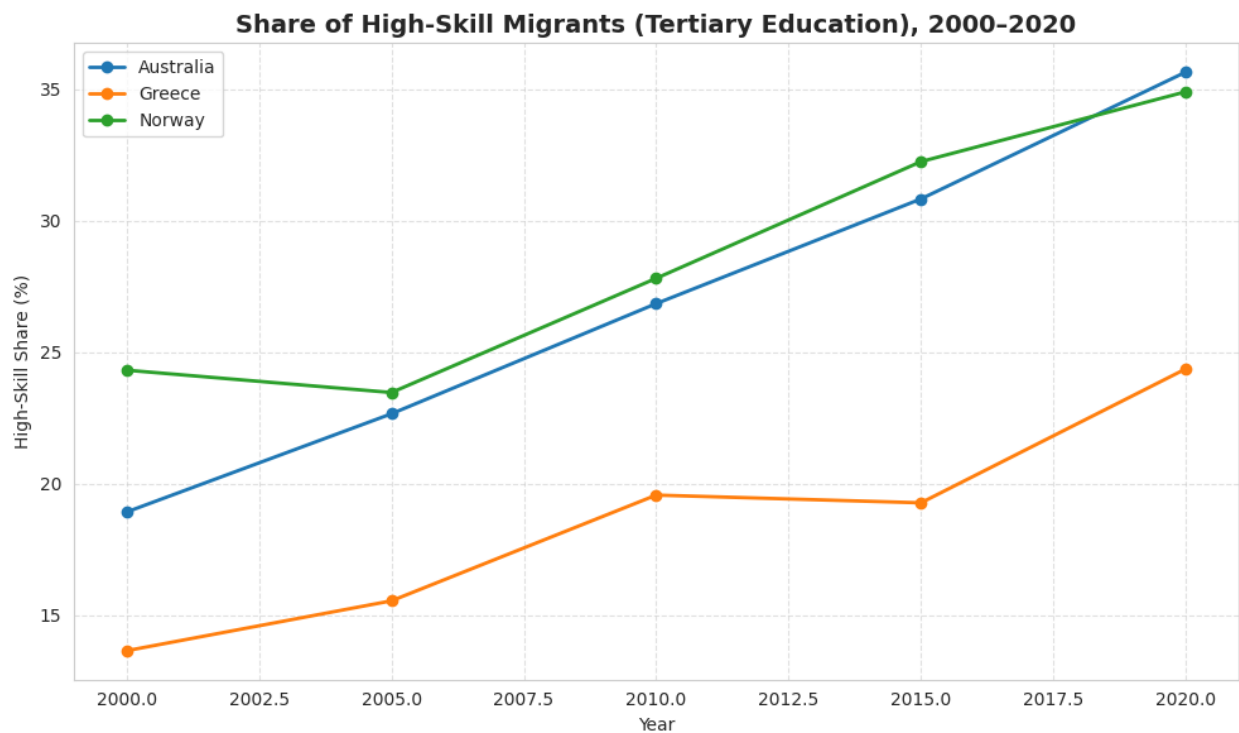


Figure 5 Share of high-skill migrants (tertiary education) for Australia, Greece, and Norway (2000-2020).

3.2 Migrant Skill Flows and Labor Market Integration

Figure 6 shows the comparison between immigrant and native employment rates in the countries of focus.

Employment & Unemployment Rates for Immigrants vs Natives (2000-2020)

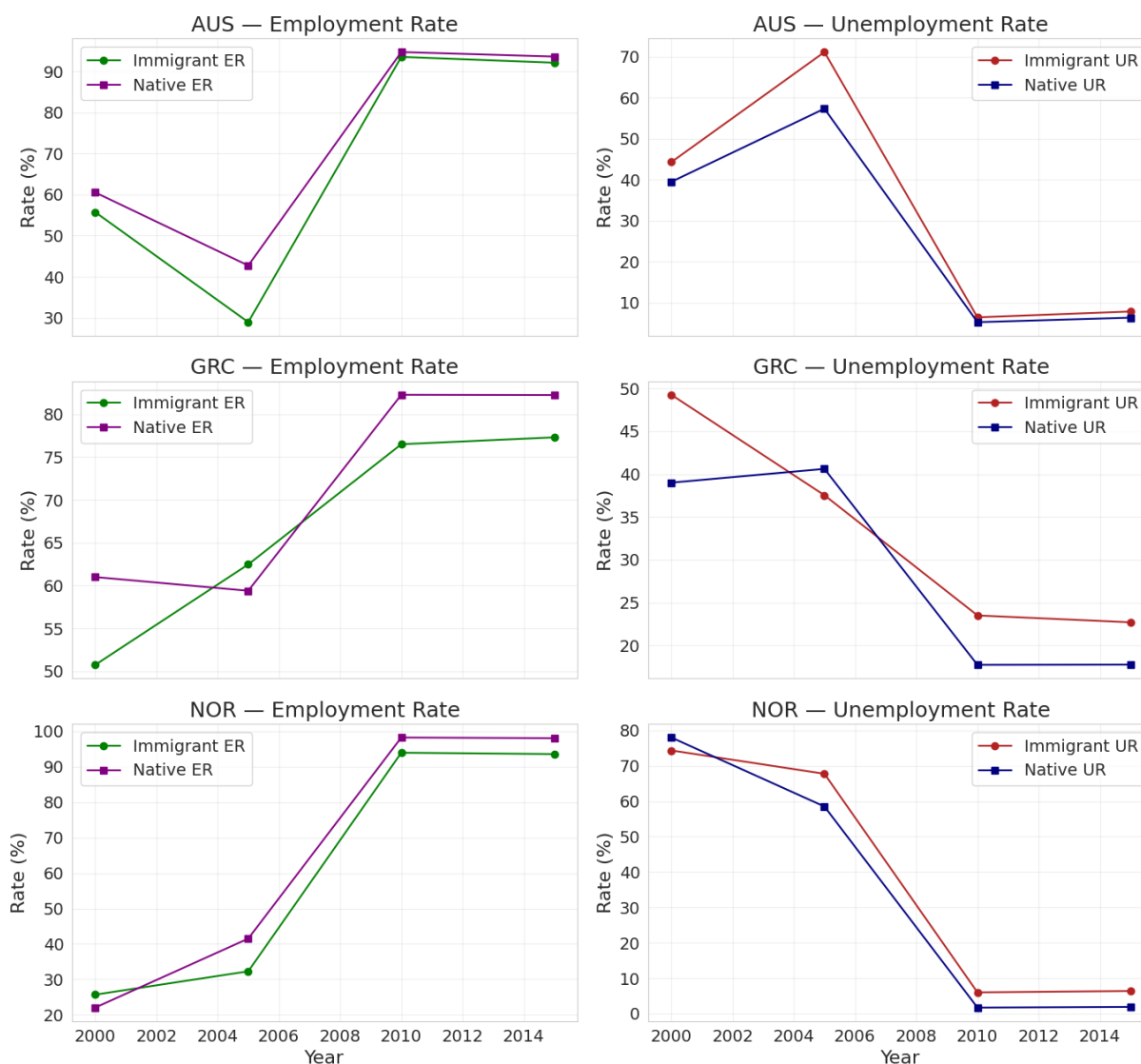


Figure 6 Employment and Unemployment rates for immigrants and natives in Australia, Greece, and Norway from 2000-2020.

3.3 Effect of Skilled Migration on Research and Development

Figure 7 plots the share of high-skill migrants against Research and Development (R&D) activity in Australia, Greece, and Norway.

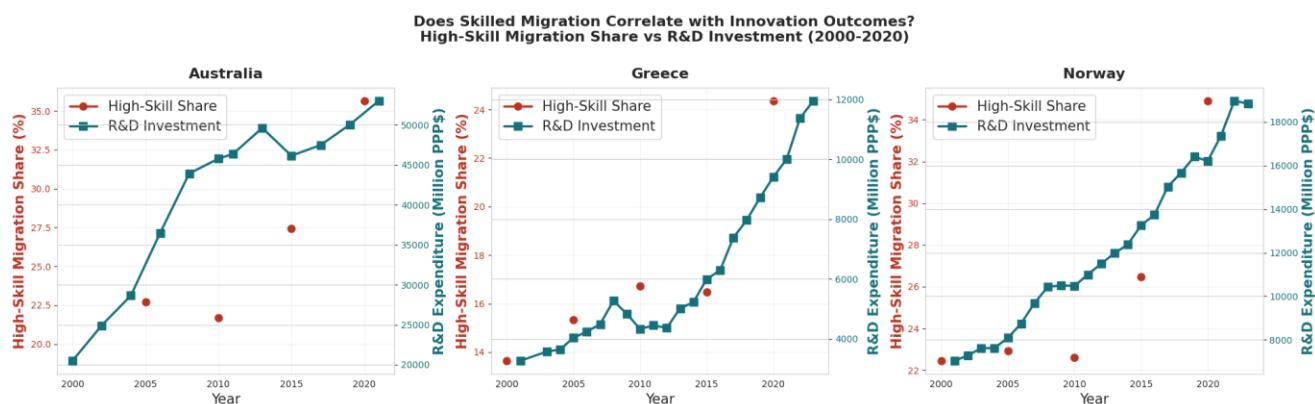


Figure 7 Plots of high-skill share and Research and Development (R&D) in Australia, Greece, and Norway.

Figure 8 shows the education levels (low, medium, or high) for both immigrants and natives in the countries of focus.

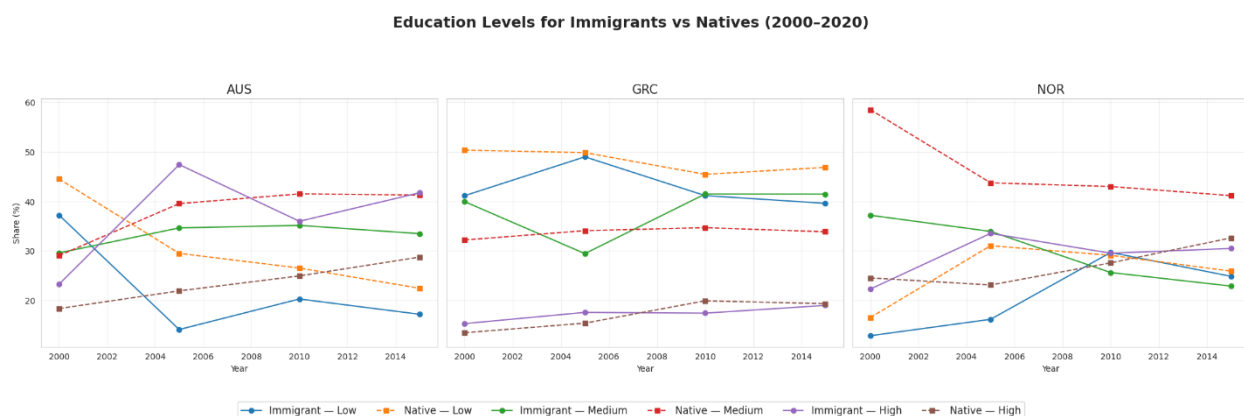


Figure 8 Education levels for immigrants and natives in Australia, Greece, and Norway (2000-2020).

4. Conclusions

The analysis of the OECD migration data from 2000-2020 highlights several patterns across Australia, Greece, and Norway.

4.1 Skill Migration Trends

High-skill migration inflows increased across the OECD, with Australia showing the most pronounced growth. Greece maintained a predominance of medium- and low-skill inflows, which Norway demonstrated balanced inflows with gradual increases in high-skill migration.

4.2 Education

Figure 8 helps explain many of the trends in the report by showing how immigrant education levels compare to those of native populations. In Australia and Norway, immigrants are increasingly more educated than natives, which supports the strong employment outcomes and growing innovation activity seen in those countries. Greece, however, shows a more mixed pattern, with immigrant education levels varying more over time and not always exceeding those of natives. This helps explain why Greece has slower labor-market improvements and why even small increases in highly educated migrants have a big effect on its R&D growth. Overall, the education differences shown in *Figure 8* help clarify why migration, employment outcomes, and innovation are linked in different ways across the three countries.

4.3 Labor Market Integration

Across Australia, Greece, and Norway, the data from 2000-2015 show a consistent pattern of improvement in labor market outcomes for both immigrants and natives. In all three countries, employment rates for immigrants rose substantially over the period and moved closer to, or even slightly above, those of natives. Overall, the trends suggest that immigrants have increasingly integrated into the labor markets of these countries, with substantial progress in closing employment and unemployment gaps relative to natives.

4.4 Innovation Outcomes

Countries with higher shares of high-skill migrants, particularly Australia, also reported stronger growth in Research and Development (R&D) investment. A correlation analysis was performed for this relationship for each of the countries of focus. Greece exhibited a statistically significant, strong positive correlation between high-skill migration and R&D investment (Pearson $r = 0.951$, $p < 0.05$). Norway showed a very strong but borderline significant correlation ($r = 0.917$, 0.083). Australia displayed a strong but not statistically significant correlation ($r = 0.757$, $p = 0.45$).

These results suggest that increases in high-skill migration are closely associated with higher R&D activity, particularly in Greece and Norway. In Australia, despite strong inflows of high-skill migrants, the statistical relationship was weaker, likely reflecting broader innovation drivers beyond migration alone.

5. Limitations

5.1 Data Consistency

The DIOC datasets vary across years in terms of variable definitions, file structures, and coverage. Harmonization was necessary, but residual inconsistencies may affect comparability. Additionally, some countries have missing observations or inconsistent reporting for certain skill groups, which may introduce bias into cross-country comparisons.

5.2 Country Selection

The focus on Australia, Greece, and Norway provides useful contrasts but does not capture the full diversity of OECD migration contexts. Results may not generalize to all member countries. These countries also differ significantly in population size, economic structure, and policy frameworks.

5.3 Indicators Used

Employment and unemployment rates, as well as R&D expenditure, are broad measures. While they offer high-level context, they do not capture more nuanced drivers of migration or economic inclusion.

5.4 Time Frame

The analysis covers 2000-2020. More recent developments, including post-2020 migration shifts and pandemic-related impacts, are not reflected.

5.5 Causality

The study describes associations between migration skill levels, labor market outcomes, and R&D activity. It does not establish causal relationships, and external factors such as policy changes or macroeconomic conditions may also play significant roles.

6. Recommendations

6.1 Strengthen High-Skill Talent Attraction

Countries with lower high-skill inflows, such as Greece, should simplify visa pathways and credential recognition to attract more educated workers. Making entry and residency processes clearer would help address labor shortages and demographic pressures. These steps would better position countries to compete for global talent.

6.2 Improve Labor-Market Integration Support

Because immigrant employment outcomes often trail those of native workers, countries should expand language training, job-matching services, and employer incentives. These supports would help newcomers enter the workforce more quickly and reduce the employment gaps observed in the results. Improving integration would also ensure better use of migrants' skills.

6.3 Connect Skilled Migration with Innovation

The link between high-skill migration and R&D activity shows the need to coordinate migration and innovation strategies. Supporting universities and firms in recruiting international talent can strengthen national innovation systems. Aligning these policy areas would help countries gain more from the skilled migrants they attract.

6.4 Improve Data Quality

Given inconsistencies in the DIOC data, countries should work toward more consistent and detailed reporting on migrant skills, employment outcomes, and education levels. Standardizing how these variables are defined and collected across years would reduce comparability issues and strengthen trend analysis. More reliable data would allow policymakers to evaluate whether integration and innovation policies are having their intended effects. In turn, this would support more informed and adaptable decision-making as migration patterns and labor-market needs continue to evolve.

7. References

OECD. (n.d.). OECD. Retrieved from <https://www.oecd.org/>

8. Appendix A: Documentation of Team Functioning

ABET Student Outcome 5: An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

Document your teamwork on Case Study 6 by answering the questions below.

Question 1: How would you rate the effectiveness of your team and its members?

The team was effective overall, and both members contributed consistently to the assigned tasks. The communication was clear throughout the project, allowing us to work efficiently.

Question 2: How did you and your team members collectively provide leadership?

Our team shared leadership depending on the task, ensuring that responsibilities were distributed fairly and progress remained steady. We shared the responsibility of guiding discussions, coordinating technical work, and clarifying analytical steps.

Question 3: How did your team members create a collaborative and inclusive environment?

To create a collaborative and inclusive environment, we regularly checked in with each other, and we made sure decisions were made collaboratively.

Question 4: How did you and your team members collectively establish goals and plan tasks?

Our team divided responsibilities based on each other's strengths and availability. We shared updates consistently to ensure tasks stayed on track.

Question 5: Why do you think that you and your team members have met the project objectives?

We met the project objectives because we communicated effectively, divided tasks and responsibilities, and checked in with each other through each stage of the project. Our final deliverables aligned with the requirements of the case study.